

Yushan Mu

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EDUCATION

University of Waterloo

Sep 2020 – Jun 2026

- Bachelor of Combinatorics and Optimization, Minor in Computing
- Courses: Object-Oriented Programming, Data Structures, Databases, Cryptography, Computer Networks

EXPERIENCE

Full Stack Developer – Versaterm

Sep 2025 – Dec 2025

- Designed and implemented an end-to-end reporting system using **Spring Boot**, **Hibernate**, and **Vue.js** to visualize employee performance metrics across internal teams, reducing manual reporting effort
- Developed **SQL** query generators to identify incomplete incident forms for **500k+** users daily
- Enabled cross-platform response template publishing to support consistent internal workflows
- Built template creation and question reordering features to improve workflow flexibility for **1200+** clients

Software Developer – Natural Resources Canada

Sep 2024 – Dec 2024

- Implemented front-end enhancements for the MapML viewer using **JavaScript**, improving user interface responsiveness and accessibility in map rendering and user interaction workflows
- Developed and maintained end-to-end tests with **Playwright** to ensure feature stability
- Built front-end prototypes to simulate API-driven user workflows using **HTML**, **CSS**, and **JavaScript**

Software Quality Analyst – Magnify Digital

Jan 2024 – Apr 2024

- Designed and executed manual test plans for new features and bug fixes using **Zephyr Scale**
- Built internal tools to interact with **REST APIs** using **JetAdmin**

IT Specialist – University of Waterloo

Sep 2022 – Apr 2026

- Diagnosed and resolved malware, driver, and network authentication issues across diverse systems
- Documented recurring issues and solutions in Confluence to improve efficiency and knowledge sharing

PROJECTS

Reliable Data Transfer Protocol | *Python, UDP, Socket Programming*

- Implemented a reliable data transfer protocol over UDP to handle packet loss, duplication, and reordering
- Designed sender/receiver programs with timeout-based retransmission and selective acknowledgements
- Built a network emulator to simulate packet loss and latency for testing

Forest Time Machine | *JavaScript, React, Node.js, boardgame.io*

- Designed and built a real-time multiplayer game with synchronized game state and turn-based mechanics
- Implemented complex game logic including phase transitions, voting systems, and card-based actions
- Built a dynamic UI that reflects game state and actions in real time across multiple phases

SKILLS

- **Programming Languages:** Python, JavaScript, Java, C, C++, SQL, HTML, CSS
- **Tools/Frameworks:** React, Node.js, Spring Boot, Hibernate, Vue.js, Playwright, Git, Bash, Jira